

# NEUROGUARDIAN CLINICAL REPORT

Explainable Seizure Detection & Neural Pattern Analysis

## 1. RECORDING SUMMARY

Filename: chb01\_01.edf  
Analysis Duration: 3600.0 seconds  
Model Type: SEIZE

## 2. DIAGNOSTIC VERDICT

!!! SEIZURE ACTIVITY FLAGGED !!!

### Clinical Interpretation:

Detected 21 clinical episodes totaling 372.5s. Pattern analysis shows high-confidence ictal rhythmic discharges.

## 3. DETECTED EPISODES LOG

| Onset Time (s) | End Time (s) | Duration (s) | Max Confidence |
|----------------|--------------|--------------|----------------|
| 172.5          | 182.5        | 10.0         | 0.8            |
| 417.5          | 422.5        | 5.0          | 0.8            |
| 525.0          | 532.5        | 7.5          | 0.8            |
| 762.5          | 772.5        | 10.0         | 0.8            |
| 922.5          | 942.5        | 20.0         | 0.8            |
| 1597.5         | 1607.5       | 10.0         | 0.8            |
| 2037.5         | 2070.0       | 32.5         | 0.9            |
| 2102.5         | 2115.0       | 12.5         | 0.8            |
| 2217.5         | 2230.0       | 12.5         | 0.9            |
| 2340.0         | 2345.0       | 5.0          | 0.8            |
| 2497.5         | 2520.0       | 22.5         | 0.8            |
| 2535.0         | 2550.0       | 15.0         | 0.8            |

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|        |        |      |     |
|--------|--------|------|-----|
| 2565.0 | 2585.0 | 20.0 | 0.9 |
| 2607.5 | 2617.5 | 10.0 | 0.8 |
| 2680.0 | 2707.5 | 27.5 | 0.9 |
| 2712.5 | 2747.5 | 35.0 | 0.9 |
| 2792.5 | 2847.5 | 55.0 | 0.9 |
| 3260.0 | 3277.5 | 17.5 | 0.9 |
| 3390.0 | 3410.0 | 20.0 | 0.8 |
| 3412.5 | 3430.0 | 17.5 | 0.8 |
| 3445.0 | 3452.5 | 7.5  | 0.8 |

## 4. EXPLANABILITY (XAI) VALIDATION

This analysis is accompanied by persistence and faithfulness metrics to ensure model trust. Cross-dataset stability is monitored to flag potential domain-shift errors.

Faithfulness Score:

High (verified via perturbation testing)

DISCLAIMER: This is an AI-assisted diagnostic tool designed for research purposes. Final clinical decisions must be made by a qualified neurologist after manual review of the raw EEG signals.